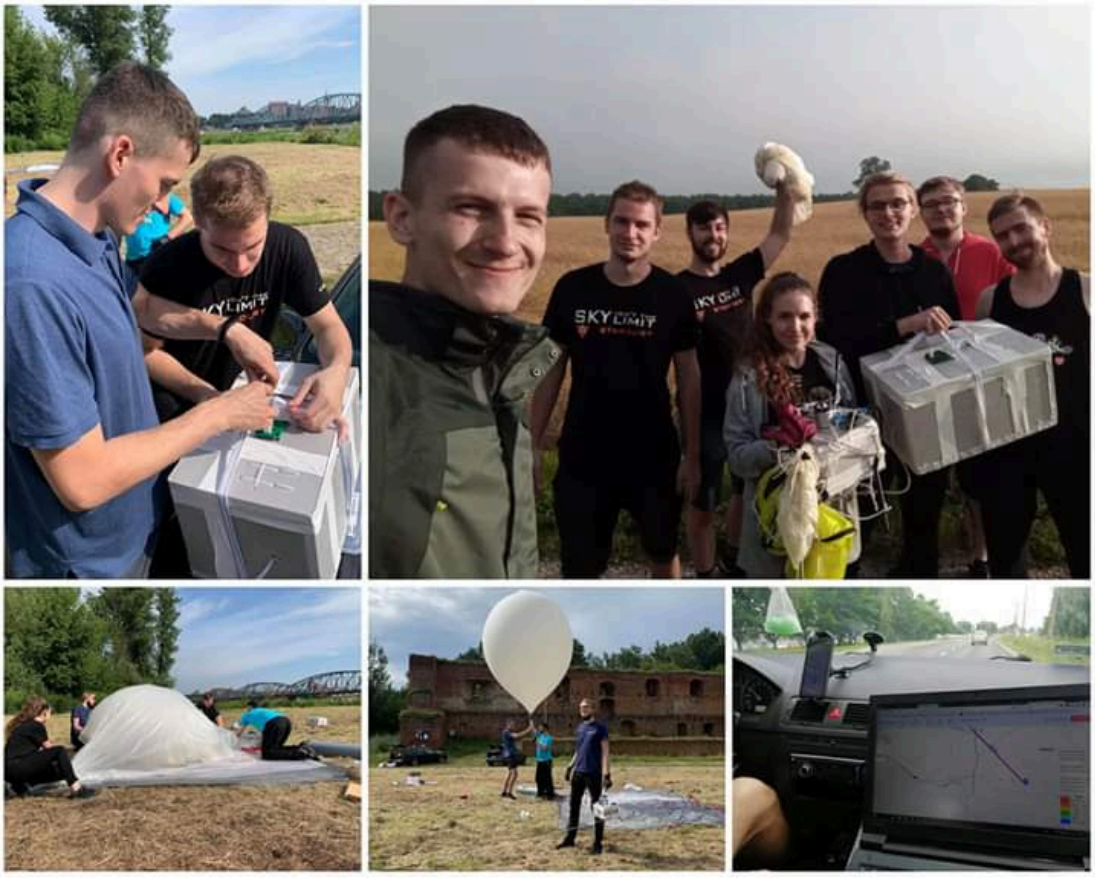
STARDUST

Test Mission Technical Report



2020/07/20

Flight Train:

The flight train consisted of a latex balloon, a parachute and two gondolas: main communications gondola and the experiment gondola. The comm gondola hosted an APRS transmitter, LoRa transceiver, WiFi ESP8266 server, camera, GPS module and a bank of sensors. The experiment gondola contained one half of the Stardust-BEXUS filtering system, ie. 6 servos and valves, pump, the first iteration of electronics PCBs and an additional WiFi client simulating E-Link communication.



Timeline (GMT+2):

- 9:00 - NOTAM begins.
- 11:36 - Liftoff.
- 12:19 ~11km AMSL - Sampling begins.
- 14:08 - 32km AMSL - LoRa packet received by a ground station in Munich, Germany (755km LoS).
- 14:33 - 33km AMSL - Raising ends; No burst.
- 16:08 - 33km AMSL - Burst; Falling starts.
- 16:14 - 16km AMSL - Last LoRa packet received.
- 16:21 - 8km AMSL - Sampling ends.
- 16:32 - Landing.
- 18:49 - Balloon recovered.

Flight path on LoRa TTN

The liftoff was conducted from Toruń, Poland.

The initial launch plan assumed a NOTAM timeline from 9:00 to 15:00. The flight train was assembled and ready for liftoff at 10:00, but the launch had to be postponed due to issues with buoyancy. The helium bottle didn't contain the advertised amount of gas, and in turn the tanked balloon didn't provide enough lifting force for our experiment.

The balloon successfully lifted off at 11:36, after being tanked with $1m^3$ more of gas.

We expected the balloon to burst after reaching 33km altitude, but it continued to float for 2 more hours. There was no cutter on board, so we couldn't terminate the mission earlier.

Communication with the balloon was maintained throughout the mission with LoRaWAN, with the last packet being received at 16km of altitude. The final location of the landing site was acquired using the "Find my device" feature on a smartphone included on-board.

Experiment assembly:

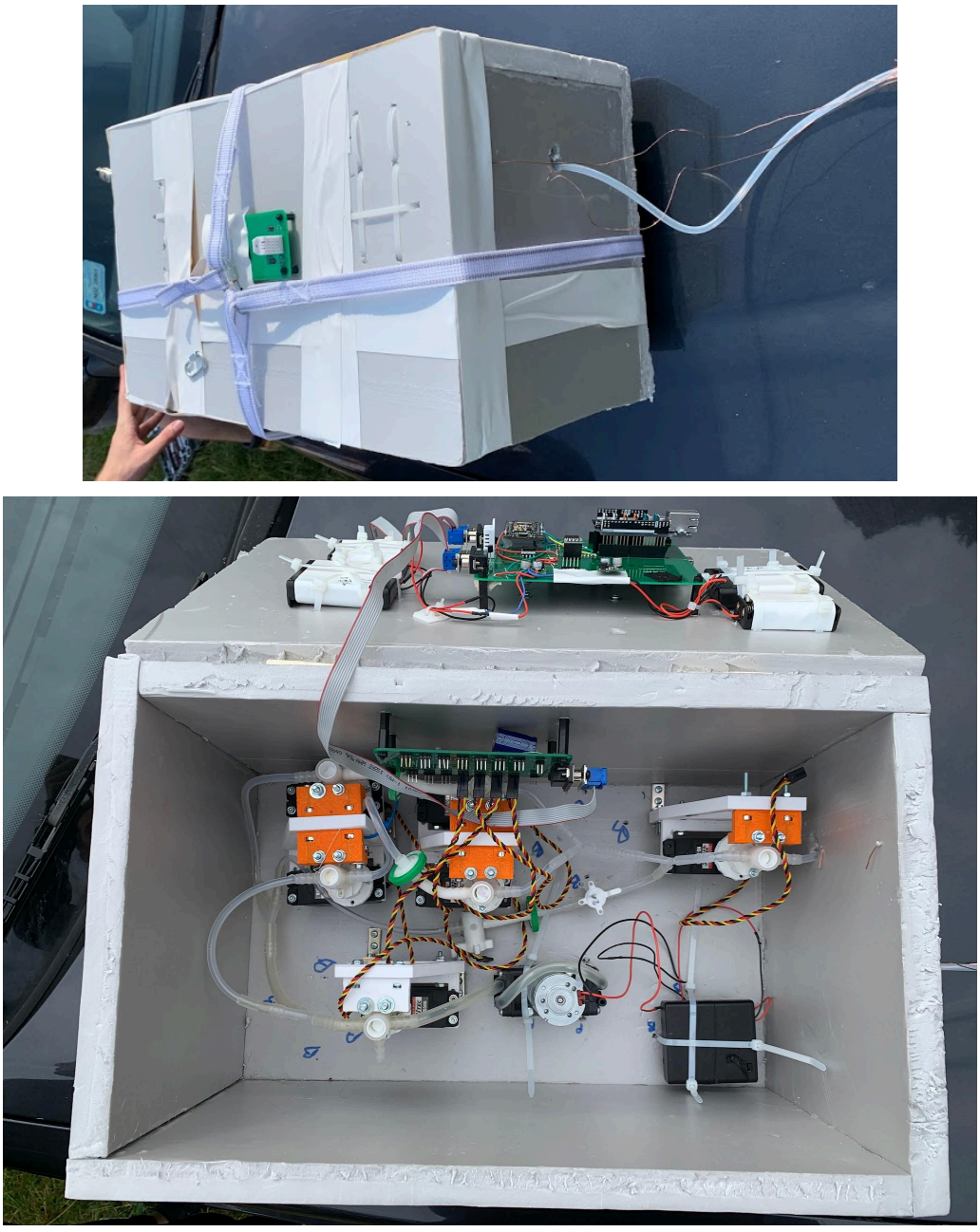


Fig. 2: Experiment assembly pre launch.



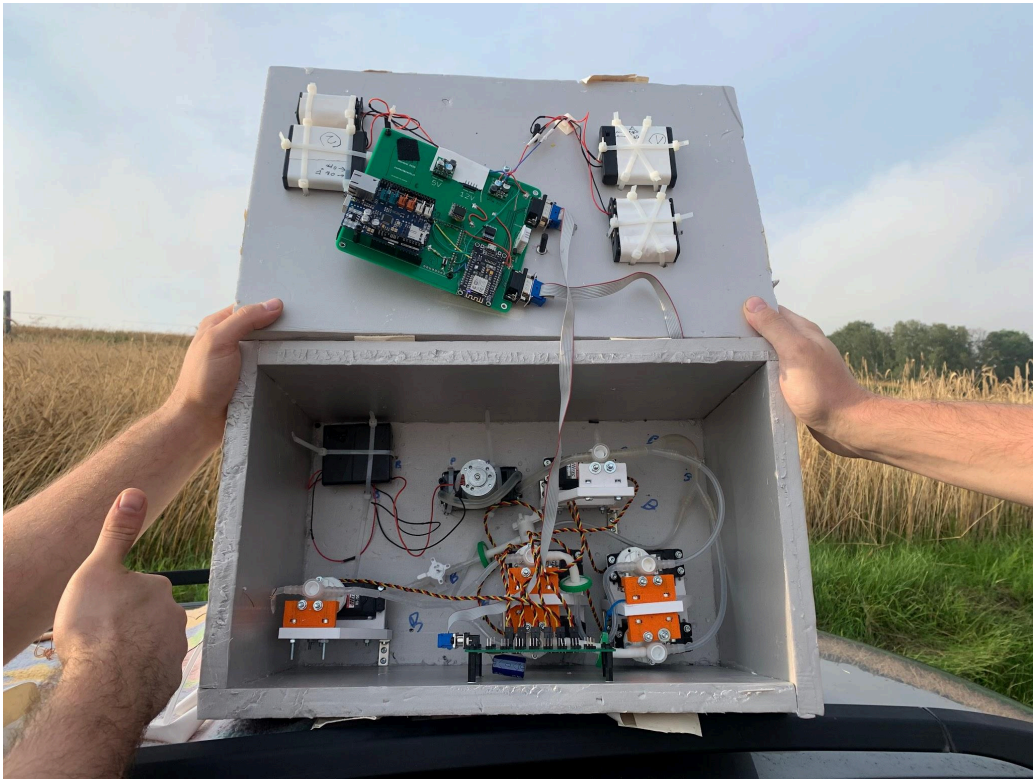


Fig. 3: Experiment assembly post launch.

Flight Data:



Fig. 4: Ground Path.

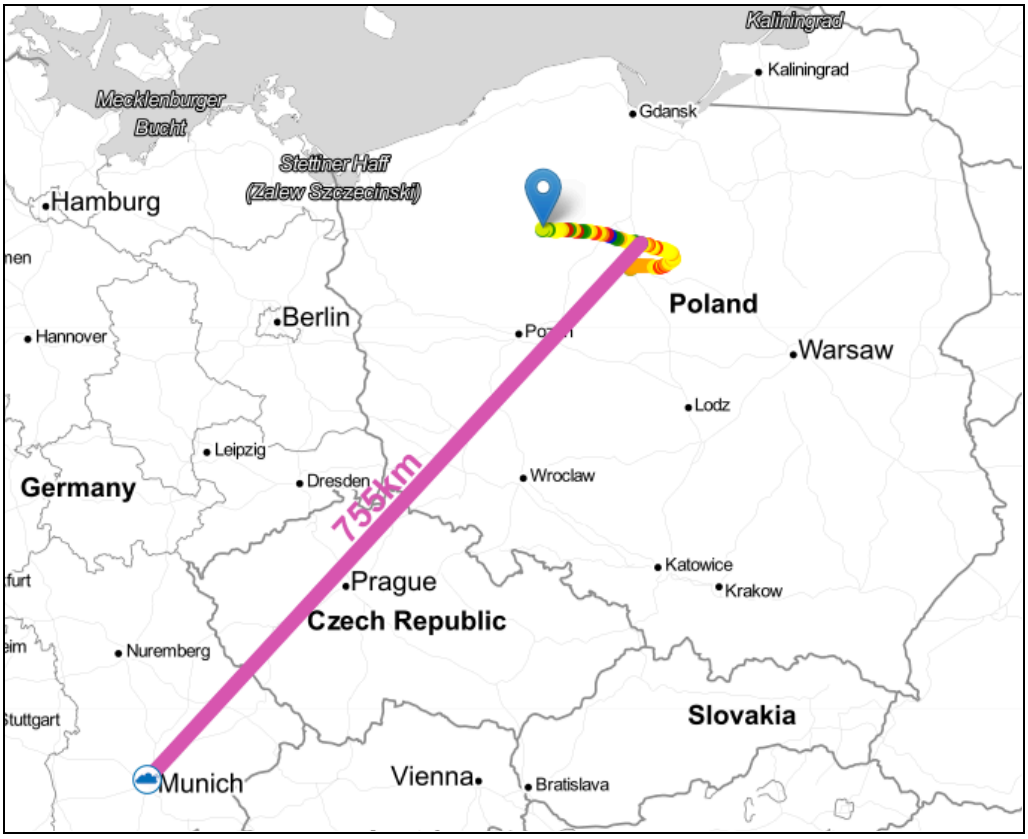


Fig. 5: Ground path and the longest packet acquisition line of sight.

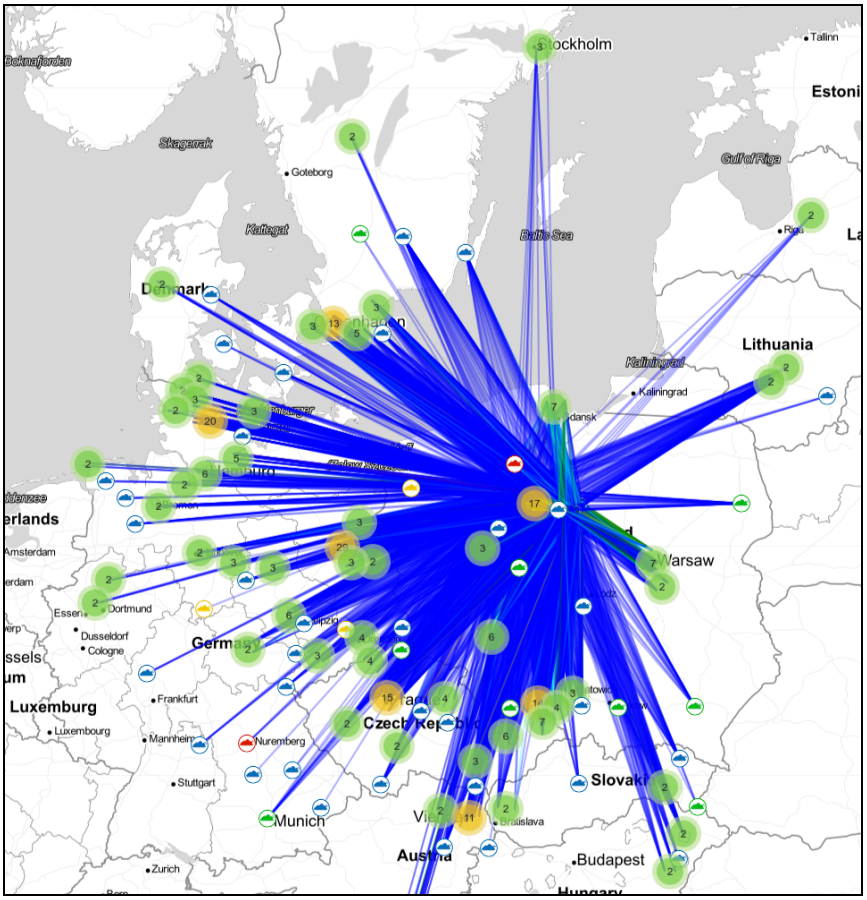


Fig. 6: All ground stations reached.

The LoRa link of 755km was a really surprising outcome of the mission. Best to our knowledge, it is one of the longest links ever achieved within the LoRa The Things Network. The previous standing records are 766 and 832km. This experience certainly sheds light on possible future missions utilizing this and other radio technology.

Sensor Data:

Two identical banks of sensors (pressure, humidity, temperature) were placed inside and outside of the experiment gondola to measure the atmospheric condition. The sensors used were: SSCMR 015PDAA5[1] for pressure and DHT22[2] for humidity, together with their built-in temperature sensors. No additional DS18B20 sensors were used. The data acquisition rate was 1Hz for all measurements.

Blue - Data from sensors inside of the experiment gondola;
Red - Data from sensors outside of the experiment gondola;

[1] https://sensing.honeywell.com/honeywell-sensing-trustability-board-mount-pressure-sensors-tsc-nsc-datasheet-50064366-d-en.pdf
[2] https://www.sparkfun.com/datasheets/Sensors/Temperature/DHT22.pdf

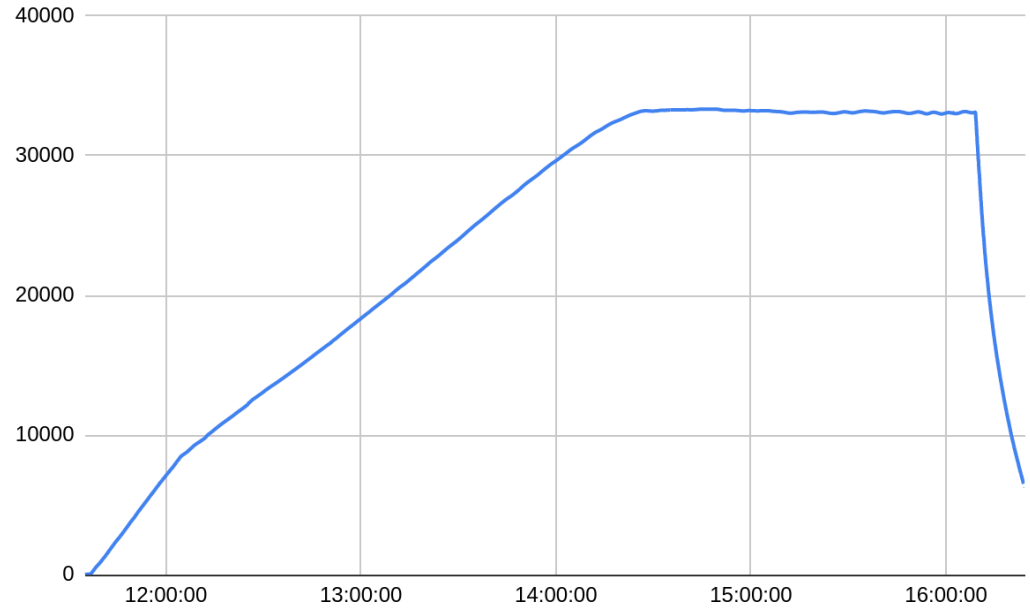


Fig. 7: GPS based flight profile [m]

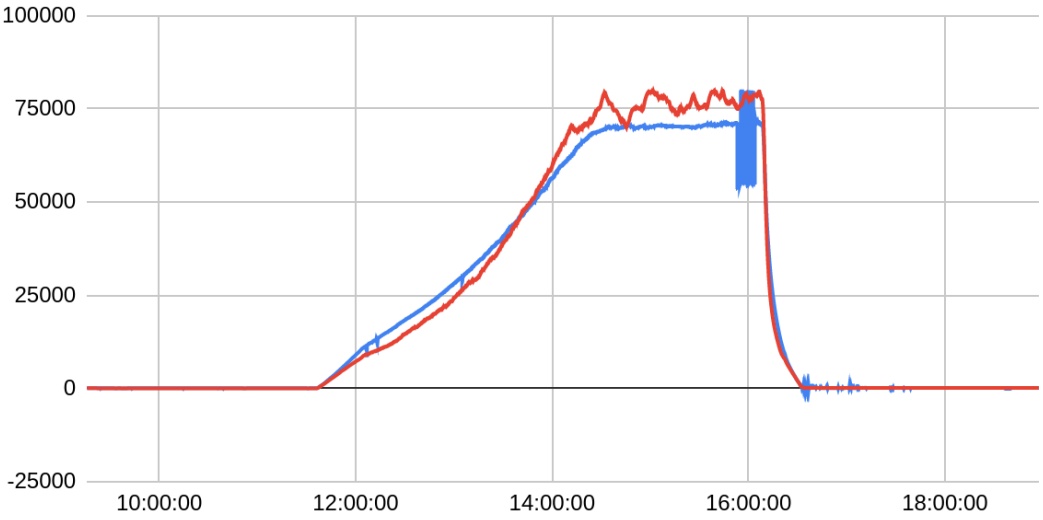


Fig. 8: Pressure and temperature based altitude calculation [m].

The altitude calculation was based on average pressure reading and a direct temperature reading. Due to the direct sampling of temperature, the calculation was prone to temperature swings outside of the gondola. The measurement based on sensors placed inside of the gondola is smoother, but was prone to error of the temperature sensor itself - value of which was not sanitized, and error value of -50*C was used in the calculation.

Due to the characteristic of the used formula and inaccurate altitude calculation, the sampling began at a proportionally lower altitude than expected. The plan was to start sampling at 15km and finish at 11km when falling to avoid sampling “dirty” air from commercial jets flying in tropopause. The actual values were 11km and 8km.

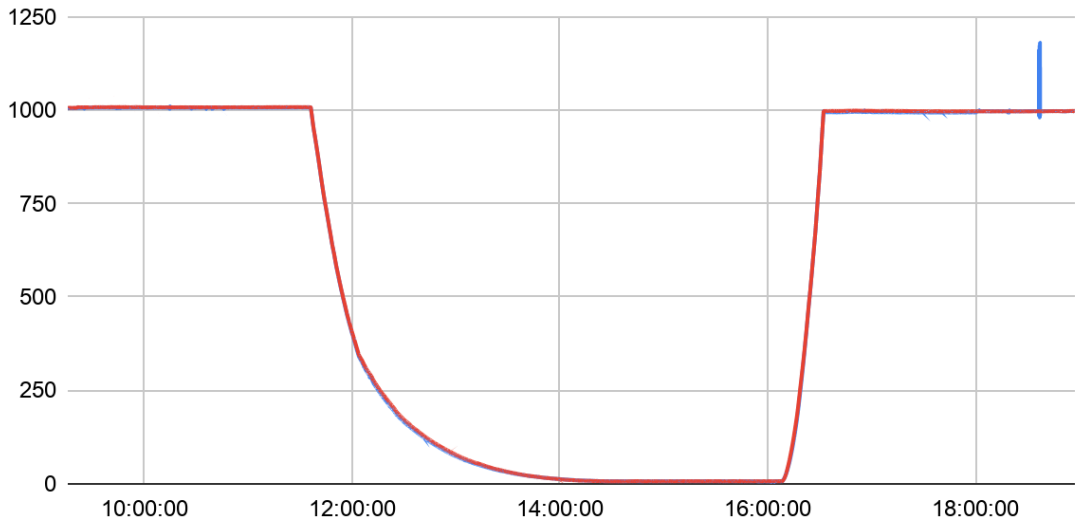


Fig. 9: Pressure readings [hPa].

Lowest pressure: 649 Pa inside, 673 Pa outside (30 measurement rolling avg.).
The pressure gradient in particular will be useful when calibrating the pressure sensors in future missions.
In addition the data shows that the BME280 sensor, recommended at CDR as a replacement for our sensor suite, is insufficient due to the pressure falling below the datasheet value of 300hPa[3].

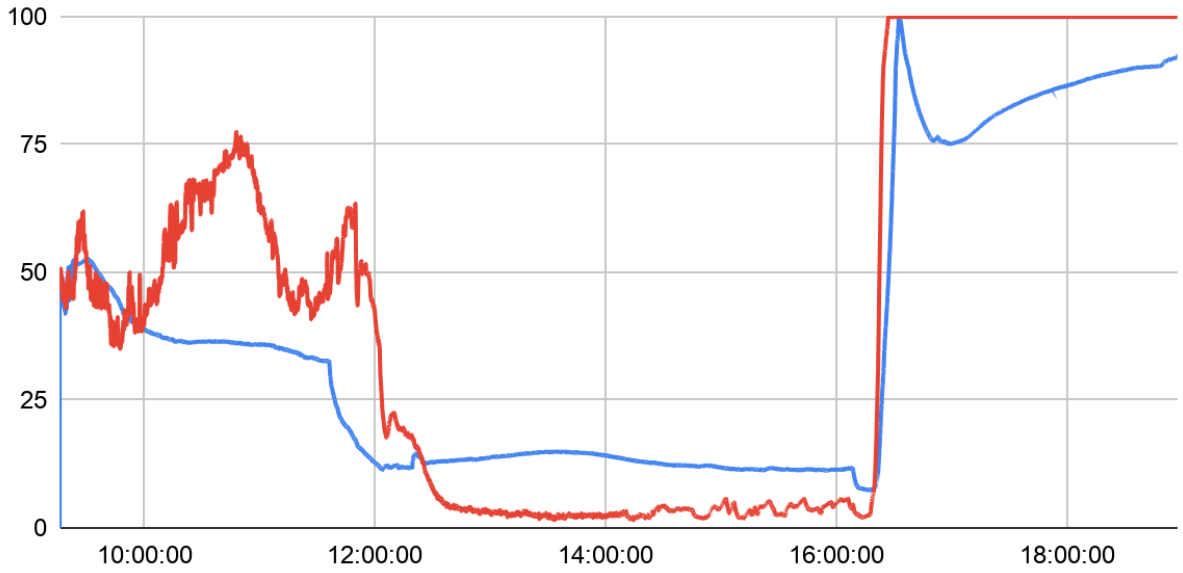


Fig. 10: Humidity readings [%RH].

[3] https://ae-bst.resource.bosch.com/media/_tech/media/datasheets/BST-BME280-DS002.pdf

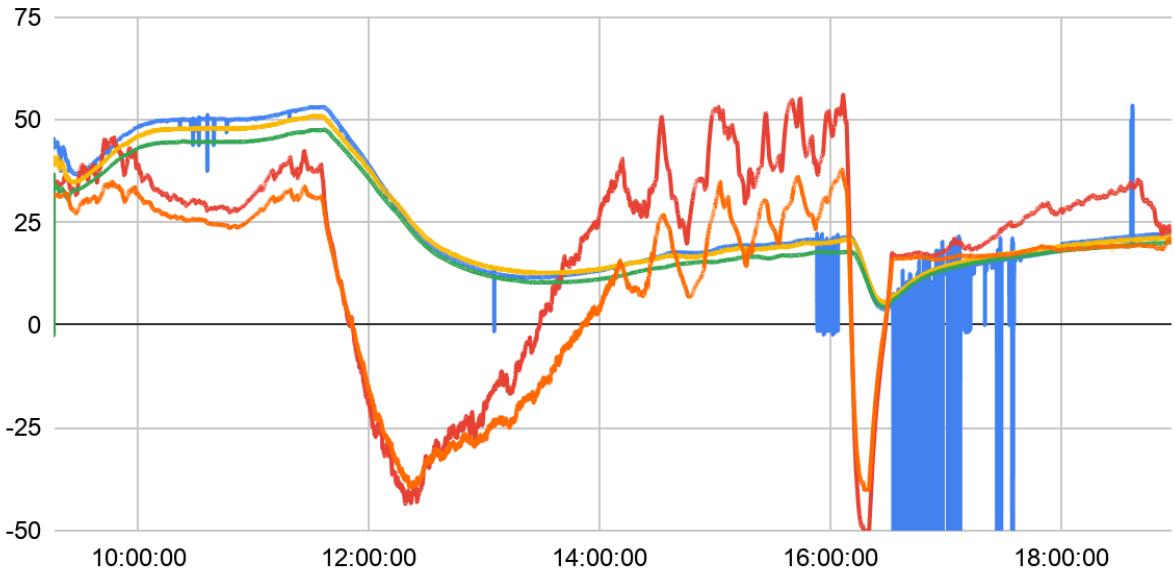


Fig. 11: Temperature readings [℃].

Lowest temperature: -40*C in flight, -50*C when falling (values limited by sensor range).

One of the temperature sensors (the one used for primary altitude calculation) encountered an issue and started returning erroneous values of 0 and -50 degrees.

The raw data from 2 different types of temperature sensors placed outside of the gondola is not scaled the same way, but shows the same characteristics of temperature.

Postmortem:

- WiFi communication used to receive GPS localization from the main gondola failed. The experiment had to rely on pressure to determine its altitude.
- APRS transmitter failed, due to an incorrectly assembled antenna.
- Formula used to calculate altitude based on pressure is insufficient at altitudes above 9km where the pressure gradient stops being linear.
- The sampling operation began and finished at lower altitude than expected, due to inaccurate information about altitude.
- The altitude estimation based on formula was prone to temperature sensor errors.
- Camera battery died approx. 30 minutes into the flight.
- The SD cards in the experiment and main gondolas were initially thought to be dead, but the data was successfully accessed the following days.
- The vacuum pump battery was drained when the gondola was found; unknown when the battery died.
- The communications gondola was additionally insulated using 3M insulate material, but it was overheated just by the heat created by the electronics alone. This information will be used when implementing a heating system for BEXUS.
- Even without insulation and with high internal volume, the temperature inside of the experiment gondola never fell below 0*C.
- Stand-offs used to mount main electronics PCBs to the ceiling of the experiment gondola compartment gave way at landing.

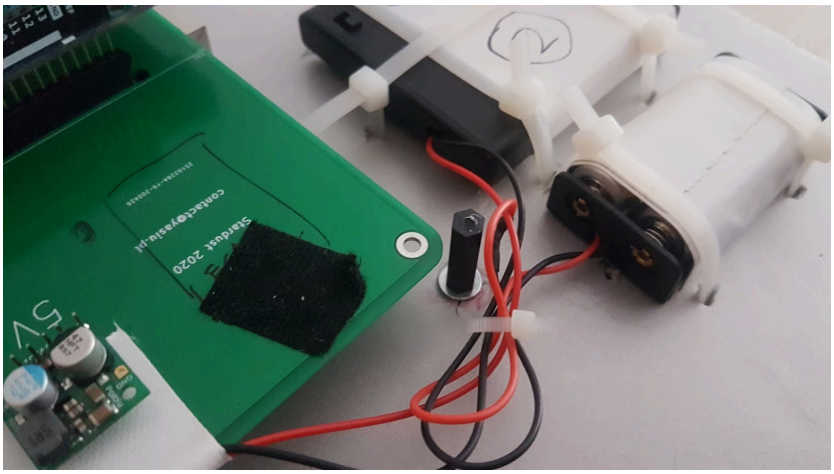


Fig. 12: Broken standoff.

Lessons Learned:

- Use a better camera that can be powered/charged from an external battery.
- Each component should either have a dedicated person assembling it, or the assembly procedure should be well documented to avoid mistakes.
- The formula used to determine altitude should have been tested better before liftoff to ensure a good estimate.
- The parameters used in the altitude formula should be sanitized better.
- In case there is no good formula to estimate altitude > 9km, either use a different formula for that range, or use predetermined pressure values to begin and end the experiment.
- Using one GPS receiver for both gondolas, and communicating the location information via WiFi is tricky. Either replace it with cable (mass permitting) or use two GPS receivers.
- Using WiFi for communication within the gondola sounds great in theory. It saves some mass of the cable and provides means to debug the experiment while still on ground, but the added complexity of operation and testing is not worth it. Additionally it could be prone to radiation at high altitudes (although this hypothesis could not be verified in this launch).
- Add more status LEDs to save yourself from stress of “is it working or not?”.
- Electronics are hot, there is probably not much need to additionally heat and/or insulate.

Conclusion

Despite some minor issues, the mission helped us to further understand the challenges of sampling microorganisms from the stratosphere. There were no major bugs in the software, electronics and mechanical design itself, and instead we found that some of the assumptions about measuring altitude from pressure were wrong. Despite the issue, the data will be useful in further missions to calibrate sensors and test assumptions. We believe that if the GPS data was available during flight, we would have better scientific data from the sampling system, due to not catching any material from tropopause and below. Overall, the self-assessment considers the mission to be a moderate success.

References

1. Raw data and mission source code on GitHub: https://github.com/simle-stardust/Stardust/releases/tag/1.0.0
2. PCB design used in the mission on GitHub: https://github.com/simle-stardust/PCB/releases/tag/1.0.0
3. WiFi connectivity source code on GitHub: https://github.com/simle-stardust/Hermes/releases/tag/1.0.0
4. Flight data on LoRa TTNMapper: https://ttnmapper.org/track/?experiment=Simle_Stardust&startdate=2020-07-20
5. Detailed photo album on Google Photos: https://photos.app.goo.gl/5xPo44yjAZN1kjAg8
6. Project page: https://stardust-bexus.net
7. SimLE science club page: https://simle.pl